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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE PROJECT (24AD3107)

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TITLE OF THE PROJECT	Sentiment Analysis on Product/Movie Reviews	
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Abstract

Sentiment Analysis on Product/Movie Reviews is an NLP-based system designed to automatically understand and classify opinions expressed in user reviews. The system analyzes review text and identifies whether the sentiment is positive, negative, or neutral, helping businesses and platforms understand customer and audience feedback efficiently. The proposed approach follows an NLP pipeline that includes text preprocessing, tokenization, feature extraction, and machine learning-based classification. Initially, the collected reviews are cleaned by removing unnecessary characters, punctuation, and common words. The text is then tokenized into individual words to make it suitable for further processing. Term Frequency-Inverse Document Frequency (TF-IDF) is used to convert the processed text into numerical feature vectors by identifying the importance of words within the reviews. These features are then used to train machine learning models such as Naïve Bayes and Logistic Regression for sentiment classification.

The performance of both models is evaluated using metrics such as accuracy, precision, recall, and F1 score to determine their effectiveness in classifying reviews. The system can analyze a large number of reviews and provide useful insights without requiring manual analysis. The generated sentiment results can help businesses identify customer satisfaction, understand common complaints, and make improvements based on user feedback. By combining NLP techniques with machine learning, the proposed system provides an efficient and practical solution for extracting meaningful sentiment from unstructured review data and supports better understanding of user opinions and decision-making.

Signature of the Guide HOD